H4 Premium/Discount Anchor Audit Report

Date: 2026-05-24
Auditor: Automated code analysis
Scope: Full comparison of H4 P&D logic between v17.25 (predecessor) and v17.54 (current)
Method: `grep`, `diff`, `sed` — exact line-by-line extraction and comparison

> **NOTE**: No file named `v17.53` exists in the repo. The immediate predecessor to v17.54 is **v17.25**. Git history confirms: `afaf11b feat(v17.54): upgrade entire stack from v17.25 to v17.54`. All comparisons are v17.25 ↔ v17.54.

Section 1: H4 P&D Code Comparison (Side-by-Side)

1.1 H4 Range Source — `request.security()` Calls

Component	v17.25 Line	v17.54 Line	Code	Identical?
H4 Range High	740	740	`float h4RangeHighRaw = request.security(syminfo.tickerid, "240", ta.highest(high, effectiveLookback), lookahead=barmerge.lookahead_off)`	✅ YES
H4 Range Low	741	741	`float h4RangeLowRaw = request.security(syminfo.tickerid, "240", ta.lowest(low, effectiveLookback), lookahead=barmerge.lookahead_off)`	✅ YES

1.2 Range Compression / Clamping Logic

Component	v17.25 Lines	v17.54 Lines	Identical?
Raw zone height calc	752	752	✅ YES
Initial assignment	756–757	756–757	✅ YES
Compressed range override	764–765	764–765	✅ YES
Midpoint-based clamping	768–771	768–771	✅ YES
Min height expansion	777–781	777–781	✅ YES

1.3 Equilibrium (EQ) Calculation

Component	v17.25 Line	v17.54 Line	Code	Identical?
h4Eq	784	784	`h4Eq = (h4RangeHigh + h4RangeLow) / 2.0`	✅ YES
h4Range	785	785	`h4Range = h4RangeHigh - h4RangeLow`	✅ YES

1.4 Premium/Discount Threshold Definitions

Variable	v17.25 Line	v17.54 Line	Code	Identical?
h4PremiumTop	795	795	`h4PremiumTop = h4RangeHigh`	✅ YES
h4PremiumBottom	796	796	`h4PremiumBottom = h4Eq + (h4Range * neutralZoneWidth / 200.0)`	✅ YES
h4DiscountTop	797	797	`h4DiscountTop = h4Eq - (h4Range * neutralZoneWidth / 200.0)`	✅ YES
h4DiscountBottom	798	798	`h4DiscountBottom = h4RangeLow`	✅ YES
neutralZoneWidth input	590	590	`neutralZoneWidth = input.float(33.3, ...)`	✅ YES

1.5 Bias Classification

Variable	v17.25 Line	v17.54 Line	Code	Identical?
h4Close	801	801	`h4Close = request.security(syminfo.tickerid, "240", close, lookahead=barmerge.lookahead_off)`	✅ YES
h4Bias	805	805	`h4Bias = h4Close > h4PremiumBottom ? 1 : h4Close < h4DiscountTop ? -1 : 0`	✅ YES
inPremiumZone	809	809	`inPremiumZone = close > h4PremiumBottom`	✅ YES
inDiscountZone	810	810	`inDiscountZone = close < h4DiscountTop`	✅ YES
inEquilibriumZone	811	811	`inEquilibriumZone = not inPremiumZone and not inDiscountZone`	✅ YES
pdZone	812	812	`pdZone = inPremiumZone ? 1 : inDiscountZone ? 0 : -1`	✅ YES
h4ZoneStr	815	815	`string h4ZoneStr = inPremiumZone ? "PZ (red)" : inDiscountZone ? "DZ (green)" : "EQ (gray)"`	✅ YES
h4BiasNeutral	825	825		✅ YES

Variable	v17.25 Line	v17.54 Line	Code	Identical?
			`bool h4BiasNeutral = h4Bias == 0`	
h4BiasBullish	826	826	`bool h4BiasBullish = h4Bias == 1`	✅ YES
h4BiasBearish	827	827	`bool h4BiasBearish = h4Bias == -1`	✅ YES
h4FibLevel	830	830	`h4FibLevel = h4Range > 0 ? (close - h4RangeLow) / h4Range * 100 : 50.0`	✅ YES

1.6 Bias Alignment & Counter-Trend Logic

Variable	v17.25 Line	v17.54 Line	Code	Identical?
h4BiasChanged	975	975	`h4BiasChanged = ta.change(h4Bias) != 0`	✅ YES
biasAlignedLong	981	981	`biasAlignedLong = h4Bias == 1 or h4Bias == 0`	✅ YES
biasAlignedShort	982	982	`biasAlignedShort = h4Bias == -1 or h4Bias == 0`	✅ YES
isCounterTrendLong	988	988	`isCounterTrendLong = h4Bias == -1`	✅ YES
isCounterTrendShort	989	989	`isCounterTrendShort = h4Bias == 1`	✅ YES

Section 2: Change Analysis

Verified diff command:

```
diff <(sed -n '590p;740,830p;975,989p' v17.25.pine) \
     <(sed -n '590p;740,830p;975,989p' v17.54.pine)
# Result: NO OUTPUT (files identical in these ranges)
```

Comprehensive logic block diff:

```
diff <(sed -n '1p;590p;600,700p;740,830p;975,1000p;1100,1200p;1200,1500p;1500,1600p' v17.25.pine) \
     <(sed -n '1p;590p;600,700p;740,830p;975,1000p;1100,1200p;1200,1500p;1500,1600p' v17.54.pine)
# Result: NO OUTPUT (files identical in these ranges)
```

Summary of ALL code differences between v17.25 and v17.54

Only **3 categories** of changes exist between the two files:

Category	Location	Nature
1. Version string substitutions	~30 comment lines scattered throughout	`"v17.25"` → `"v17.54"` in comments and HUD table cell
2. S&D zone rendering	Lines 2035–2137 (v17.54)	Scalar → array-based zone tracking (visual only, see Section 3)
3. Alert architecture	Lines 2463–2528 (v17.54)	`alertcondition()` message format changed + `alert()` calls added

The entire H4 P&D calculation engine (lines 590, 740–830, 975–989, 1100–1600) is byte-for-byte identical.

Section 3: Dependency Map — What Uses H4 P&D Variables

3.1 Complete Variable Usage Catalog

`h4RangeHigh` / `h4RangeLow`

Line (both versions)	Usage	Category
740–741	Defined via `request.security()`	**Definition**
752, 756–757, 764–771, 777–781	Range compression/clamping	**Calculation**
784–785	Input to `h4Eq` and `h4Range`	**Calculation**
2029	`pdPremiumBox` top coordinate	**Visual only**
2030	`pdDiscountBox` bottom coordinate	**Visual only**
2125–2126 (v17.25) / 2165–2166 (v17.54)	EQ line rendering	**Visual only**
2659 (v17.25) / 2705 (v17.54)	Debug table display	**Visual only**

`h4Eq`

Line (both versions)	Usage	Category
784	Defined as `(h4RangeHigh + h4RangeLow) / 2.0`	**Definition**
796	Input to `h4PremiumBottom`	**Calculation**
797	Input to `h4DiscountTop`	**Calculation**
1239–1240	Guardian EQ high/low boundaries	**Filtering** (Guardian validation)
1479–1480	P&D shift detection (`pdShiftAgainstLong/Short`)	**Filtering** (setup invalidation)

`h4Bias`

Line (both versions)	Usage	Category
805	Defined from `h4Close` vs thresholds	**Definition**
975	`h4BiasChanged` detection	**Filtering**
981–982	`biasAlignedLong/Short`	**Filtering** (alert prerequisite)
988–989	`isCounterTrendLong/Short`	**Filtering** (counter-trend gate)
1141	POI score bonus	**Scoring**
1159	State machine reset on bias change	**Filtering**
1177, 1181	IPM sweep direction check	**Filtering**
1311, 1315	IPM induce sweep + bias gate	**Filtering**
1395–1397	`alignmentOK_long/short`, `alignmentConflict`	**Filtering** (critical alert gate)
1449	Guardian code assignment	**Filtering**
1467	Guardian waiting condition	**Filtering**
1553–1554	Counter-trend signal raw conditions	**Alert firing**
1785, 1821, 1826, 1832, 1855	Action plan text	**Visual only** (HUD)
2166 (v17.25) / 2206 (v17.54)	`plot(h4Bias, ...)` for webhook	**Alert payload data**
2266, 2272 (v17.25) / 2306, 2312 (v17.54)	Display ID generation	**Visual only** (labels)
2594 (v17.25) / 2640 (v17.54)	HUD table cell	**Visual only**

`pdZone` / `inPremiumZone` / `inDiscountZone` / `inEquilibriumZone`

Line (both versions)	Usage	Category
809–812	Defined from `close` vs thresholds	**Definition**
1804	`inCorrectZoneForBias`	**Visual only** (action plan)
1861, 1864	Action plan text decisions	**Visual only** (HUD)
2029–2033	P&D background box rendering	**Visual only**
2167 (v17.25) / 2207 (v17.54)	`plot(pdZone, ...)` for webhook	**Alert payload data**
2331 (v17.25) / **2371** (v17.54)	`retrace_long_in_zone = inDiscountZone`	🔴 **ALERT FIRING**
2339 (v17.25) / **2379** (v17.54)	`retrace_short_in_zone = inPremiumZone`	🔴 **ALERT FIRING**
2670 (v17.25) / 2716 (v17.54)	Debug table zone booleans	**Visual only**

`h4FibLevel`

Line (both versions)	Usage	Category
830	Defined from `close` position in range	**Definition**
1976–1980	Fib warning labels	**Visual only**
2174 (v17.25) / 2214 (v17.54)	`plot(h4FibLevel, ...)` for webhook	**Alert payload data**

Section 4: Alert Impact Analysis

4.1 Sniper Long/Short Fire Conditions

```
// v17.25 line 2318 / v17.54 line 2358 — IDENTICAL
fire_sniper_long = sniper_long_action_ready       // actionStep == "READY"
             and sniper_long_confluence_ok         // confluenceScore >= threshold
             and sniper_long_stage_ok              // dtStage >= threshold
             and
sniper_long_guardian_ok            // guardLong != 5 and guardLong != 6
             and sniper_long_state_ok              // smStateLong == 0
             and sniper_long_not_duplicate         // f_alert_ready(...)
             and barstate.isconfirmed
```

H4 P&D impact on Sniper signals: **INDIRECT via `actionStep`**
- `actionStep == "READY"` is set by the action plan logic (lines ~1785–1870)
- The action plan uses `h4BiasNeutral`, `inCorrectZoneForBias`, `inEquilibriumZone`, etc.
- However, `actionStep` is not solely gated by H4 P&D — it depends on confluence, DT stage, kill zone, etc.
- **H4 P&D does NOT directly appear in any `sniper_*` component variable.**

4.2 Retrace Long/Short Fire Conditions

```
// v17.25 line 2336 / v17.54 line 2376 — IDENTICAL
fire_retrace_long = retrace_long_in_zone           // inDiscountZone  ← USES H4 P&D
                and retrace_long_mode_ok           // Guardian allows retrace/buy
                and retrace_long_confluence_ok      // confluenceScore >= threshold
                and retrace_long_not_duplicate     // f_alert_ready(...)
                and barstate.isconfirmed
```

H4 P&D impact on Retrace signals: **DIRECT**
- `retrace_long_in_zone = inDiscountZone` (line 2331/2371) — price must be below `h4DiscountTop`
- `retrace_short_in_zone = inPremiumZone` (line 2339/2379) — price must be above `h4PremiumBottom`
- **Both conditions are byte-for-byte identical between v17.25 and v17.54.**

4.3 Counter-Trend (CT Lite) Signals

```
// v17.25 line 1553 / v17.54 line 1553 — IDENTICAL
ctLiteLongRaw = h4Bias == -1                       // ← USES H4 P&D
            and structureDir == 1
            and structureStateCode >= 3
            and kzActive
            and poiScore >= 2
            and not h4Invalidated
            and not alignmentConflict               // ← USES H4 P&D (h4Bias vs struc-
tureDir)
            and (guardLong == 1 or guardLong == 3 or not enableGuardian)
```

H4 P&D impact on Counter-Trend: **DIRECT** — `h4Bias == -1` and `alignmentConflict` both use `h4Bias`.
Both conditions are byte-for-byte identical between v17.25 and v17.54.

4.4 Continuation Signals

No explicit `fire_continuation` variable exists in either version. Continuation logic is handled through the `actionStep` state machine.

4.5 Guardian Validation

```
// v17.25 lines 1239–1240 / v17.54 lines 1239–1240 — IDENTICAL
guardianEqHigh = h4Eq + (h4Range * guardianNeutralPct / 200.0)
guardianEqLow  = h4Eq - (h4Range * guardianNeutralPct / 200.0)
```

Guardian uses `h4Eq` and `h4Range` to define its neutral band. **Identical in both versions.**

4.6 Setup Invalidation

```
// v17.25 lines 1395–1397 / v17.54 lines 1395–1397 — IDENTICAL
alignmentOK_long  = h4Bias >= 0 and not h4Invalidated
alignmentOK_short = h4Bias <= 0 and not h4Invalidated
alignmentConflict = (h4Bias == 1 and structureDir == -1) or (h4Bias == -1 and struc-
tureDir == 1)

// v17.25 lines 1479–1480 / v17.54 lines 1479–1480 — IDENTICAL
pdShiftAgainstLong  = confirmed and close < h4Eq and close[1] >= h4Eq
pdShiftAgainstShort = confirmed and close > h4Eq and close[1] <= h4Eq
```

All setup invalidation logic using H4 P&D is **identical in both versions.**

4.7 TP Calculation (uses H4 P&D thresholds)

```
// v17.25 lines 1419–1420 / v17.54 lines 1419–1420 — IDENTICAL
tpLong  = tpModeInput == "Fixed R:R" ? entryLong + (entryLong - slLong) * defaultRR :
h4PremiumBottom
tpShort = tpModeInput == "Fixed R:R" ? entryShort - (slShort - entryShort) *
defaultRR : h4DiscountTop
```

When TP mode is NOT “Fixed R:R”, TP targets use `h4PremiumBottom` / `h4DiscountTop`. **Identical in both versions.**

Section 5: Conclusion

Finding: H4 P&D Anchor Logic Is 100% IDENTICAL Between v17.25 and v17.54

Every line of code in the H4 Premium/Discount calculation and filtering pipeline — from `request.security()` calls through bias classification, alignment gates, guardian validation, setup invalidation, and alert fire conditions — is **byte-for-byte identical** between v17.25 and v17.54.

What Actually Changed (3 items only)

#	Change	Affects Alert Firing?	Evidence
1	Version strings in comments/HUD (`"v17.25"` → `"v17.54"`)	❌ No	~30 comment-only substitutions
2	S&D zone rendering: scalar → array multi-zone	❌ No — **visual only**, zone boxes have zero connection to `fire_*` conditions	Lines 2035–2137 (v17.54) — `demandTop` / `supplyTop` etc. are never read by any alert logic
3	Alert message format: `{{plot_X}}` → simplified JSON + `alert()` calls added	❌ No — message format doesn't affect **when** alerts fire, only **what data** they carry	Lines 2463–2528 (v17.54)

Definitive Answer

The H4 P&D anchor system has ZERO impact on any alert firing difference between v17.25 and v17.54, because it was never changed.

If alerts are behaving differently in v17.54 vs v17.25, the cause is NOT in:
- H4 range calculation
- EQ line calculation
- Premium/Discount thresholds
- Bias classification
- Zone detection logic
- Guardian validation
- Alignment/conflict logic
- Setup invalidation
- Any fire condition formula

The only functional code changes between the two versions are:
1. **S&D zone box rendering** (visual only — no signal impact)
2. **Alert message payload format** (when alerts fire is unchanged; what they contain differs)